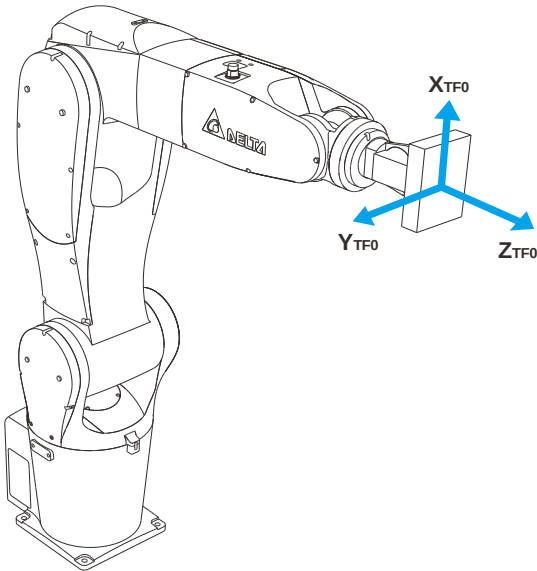


4.7.4 How to teach the Tool frame directions

Set the coordinate frame (Mode) to World (both the User Frame ID and Tool Frame ID are set to 0 in this mode), and then follow these steps to convert the default Tool Frame Z axis direction for it to align with the direction of the end effector.



Default Tool Frame directions

- 1. Click **Determine** in the Tool Orientation area to open the Tool Orientation Calculation screen.

Tool Orientation

Input :

(1) Enter the tool orientation data (Unit : Degree)

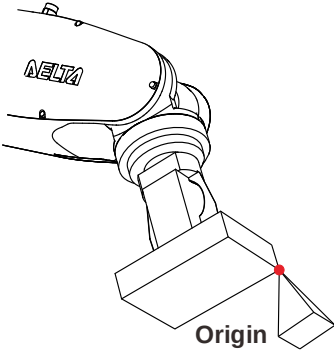
Roll (RX)	Pitch (RY)	Yaw (RZ)
0.000	0.000	0.000

(2) Determine the tool orientation data by teaching

Determine

Tool Orientation Set

- 2. Select a fixed point, and then move the end effector to the fixed point. Select the **Origin** table row, and then click **Teach** to save the current position to the row.



Tool Orientation Calculation

ID :

Input :

Enter data and Calculate

	X (mm)	Y (mm)	Z (mm)
Origin	444.001	0.000	211.321
+ X axis	0	0	0
+ Y axis	0	0	0

Teach

Tool Orientation Set

Get

Output :

Roll (RX)	Pitch (RY)	Yaw (RZ)
0	0	0